LEVEL 3 · PART I

Git & GitHub

3

Goal: Use Git and GitHub as a team collaboration and release system, not just a save button.

RECOMMENDED TIME	TOPICS
2-3 weeks	13

PRACTICE / CAPSTONE

Set up a repo with protected main, PR workflow, semantic-version tags, and a GitHub Actions workflow that runs on PRs and releases on tags.

Topics

#	Topic	What you will learn
1	Branching and Merging **p.565**	branches, fast-forward vs merge commit, conflicts
2	Rebasing **p.577**	rebase vs merge, interactive rebase, golden rule, force-with-lease
3	Cherry-picking **p.587**	hotfix backports, release branches
4	reset and revert **p.599**	soft/mixed/hard reset, revert on shared branches, reflog recovery
5	stash **p.609**	saving WIP, stash list/pop/apply, stash branches
6	Tags **p.619**	lightweight vs annotated, release tags, signing
7	bisect **p.629**	binary search for a regression, automated bisect with a test script
8	Protected Branches and Pull Requests **p.639**	branch protection rules, required reviews/checks, CODEOWNERS, merge queue
9	Trunk-based Development and GitFlow **p.651**	branching strategies, feature flags, trade-offs
10	Semantic Versioning and Release Management **p.663**	semver, conventional commits, changelogs, GitHub Releases
11	GitHub Actions: Workflows, Jobs and Triggers **p.677**	workflow YAML, events, jobs, steps, matrix, caching

#	Topic	What you will learn
12	GitHub Environments and Secrets **p.691**	environments, protection rules, secrets, OIDC to cloud
13	Artifacts and Runners **p.705**	upload/download artifacts, GitHub-hosted vs self-hosted runners
★	Level capstone project **p.719**	Set up a repo with protected main, PR workflow, semantic-version tags, and a GitHub Actions workflow that runs on PRs and releases on tags.

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.
4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow

